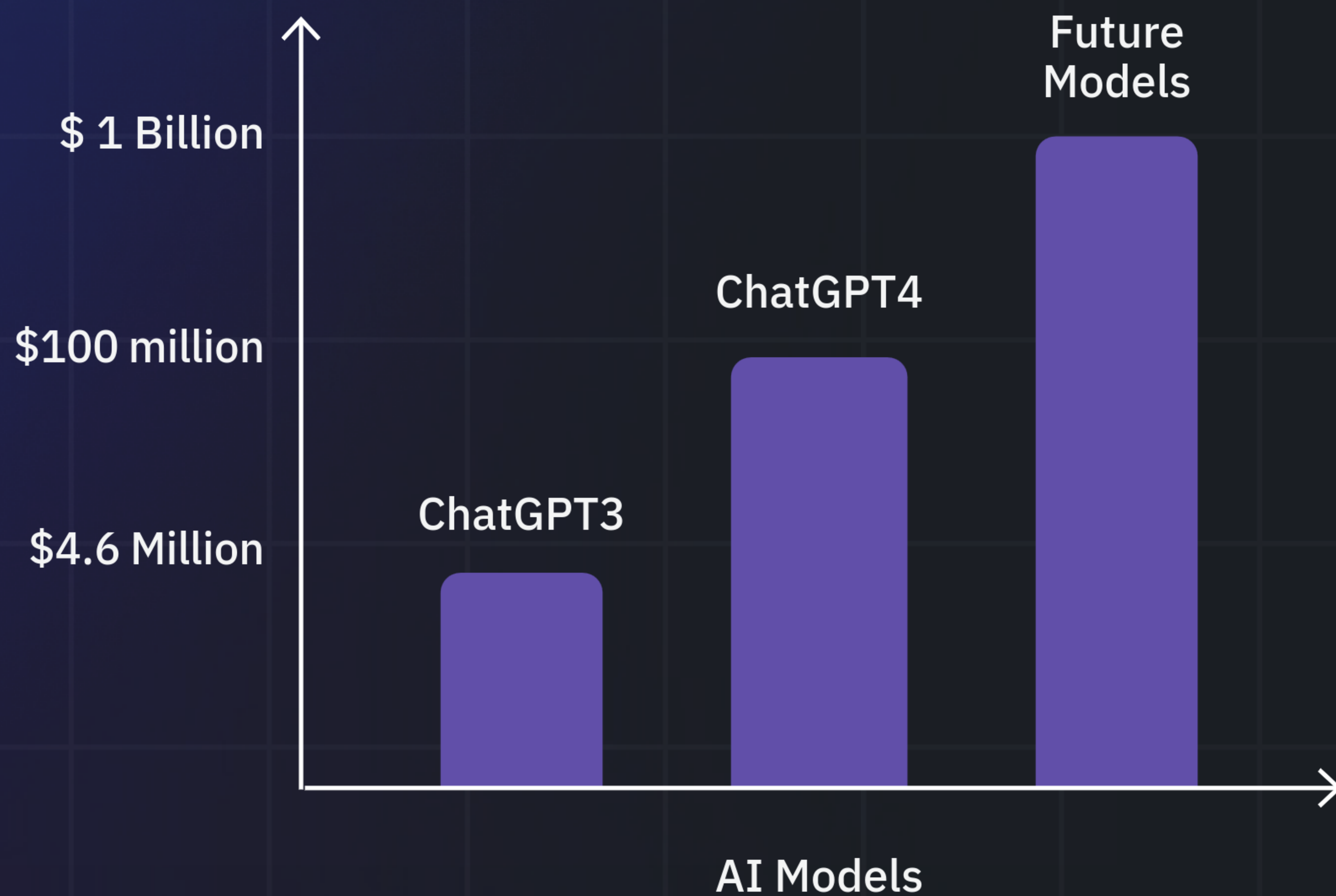


8.1.2 Cost effective

It is more efficient and cost-effective to fine-tune an existing model than to train a new one from scratch. ChatGPT-3, which was launched in 2020 and had 175 million parameters, reportedly cost \$4.6 million to train.¹ The latest version, ChatGPT-4, cost about \$100 million to build.² Anthropic CEO has stated some models currently in training will cost around a billion dollars.³



As Dario Amodei noted, you do not have a few million dollars to train an LLM model from scratch—or a billion dollars. But you can speak for everyone when you say it will be cheaper to fine-tune an existing LLM than to build a new one from scratch for personal, professional, or even business use.

<https://arxiv.org/abs/2310.06694>

8.1.3 Domain Specificity

Fine-tuning allows the LLM to understand better and generate content relevant to specific industries or topics. For instance, in economics, liquidity has two meanings- the ease of turning an asset into money and the availability of cash for spending or investment in an economy.

[1] <https://fortune.com/2024/04/04/ai-training-costs-how-much-is-too-much-openai-gpt-anthropic-microsoft/>

[2] <https://www.wired.com/story/openai-ceo-sam-altman-the-age-of-giant-ai-models-is-already-over/>

[3] <https://www.entrepreneur.com/business-news/anthropic-ceo-ai-will-cost-10-billion-to-train-by-2025/476750>